# Update on functional recovery process for the injured athlete: return to sport continuum redefined

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## INTRODUCTION

The traditional functional recovery model and return to sport (RTS) decision-making process after long-term injury is insufficient in the real world of sports medicine. There are lower than ideal numbers of athletes returning to competitions after certain injury types.[1 2] Additionally, those who do RTS have heightened risk of reinjury,[3] may not return to preinjury performance levels,[4] or may be unable to sustain the same level of competitive play in the subsequent years after injury.[5] This paper provides an update on the RTS process, reflecting the new literature and knowledge in the area as well as applied practice, to support practitioners working with athletes after injuries and in particular after severe injuries (>28 days).[6]

## TRADITIONAL REHABILITATION APPROACHES

A dichotomous conception of functional recovery was common in the past (and often still present) which involved the separation of clinical rehabilitation and RTS. Initially, the patient begins with the medical team before been transferred to the performance team (coaching and fitness staff) for sport-specific training and RTS. There is often little or no overlap in the process, and limited communication and sharing of knowledge during the functional recovery process.

When there is a need to prepare the athlete for direct re-entry into sport after injury, it is necessary to consider an overlap of the rehabilitation and RTS processes. This area of overlap requires specialised personnel who are concerned with the transition from classic rehabilitation (ie, from clinic) to RTS. This commonly takes place 'on-field' and bridges the gap between rehabilitation and sports training (figure 1).

Recently, an international consensus has recognised a newly defined RTS process, which acknowledges a continuum of three elements, emphasising a graded, criterion-based progression, which is applicable for any sport and aligned with RTS goals.[7] These include return to participation, such as modified training, but not been able to return to competitive sport; RTS, which is characterised as returning back to the same competitive sport, but not necessarily returning back to previous levels of performance; and return to performance ($RTP_{erf}$), which described the resumption of sport to a previous level at the same or higher level of performance. The goal of rehabilitation largely depends on the individual, but the achievement of maximal functional recovery possible aims to return an athlete back to his/her performance, as opposed to merely back to sport.

## TIME FOR AN UPDATED MODEL

There is a need to update the traditional functional recovery process by strengthening the RTS towards performance. As such, we have merged the new RTS process, which now leads to a more complex model (figure 2) and more clearly describes the whole functional recovery process.

This model is endowed with its complexity[8] and should be biopsychosocial,[9] in which an optimal result (ie, accelerated $RTP_{erf}$ with low reinjury risk) can only be obtained by adopting a multifactorial interdisciplinary approach, which requires teamwork between specialists of the medical and performance departments. It is essential that each step is undertaken in succession per criterion-based progression[7] and that each step be fully complete. Failure to do so can result in incomplete or failed rehabilitation (eg, early reinjury on RTS or even injury during the late-stage rehabilitation process). Key aspects of the model include the RTS progressions of (1) on-field rehabilitation, (2) return to training, (3) return to competitive match play, and (4) $RTP_{erf}$. The length of each stage will reflect the type of injury and specific context of that injury. Importantly, $RTP_{erf}$ should be confirmed as the ability to perform at the same or higher levels of performance during competitive sport, thus can only be confirmed after an athlete has actually 'returned to competition'. This approach is likely outside of the remit of most large multidisciplinary teams and requires the training of individual staff specialised in late-stage rehabilitation and RTS training. Furthermore, there is a need to have an individual in charge to manage the whole process, a case manager which is typically the specialist sports medicine

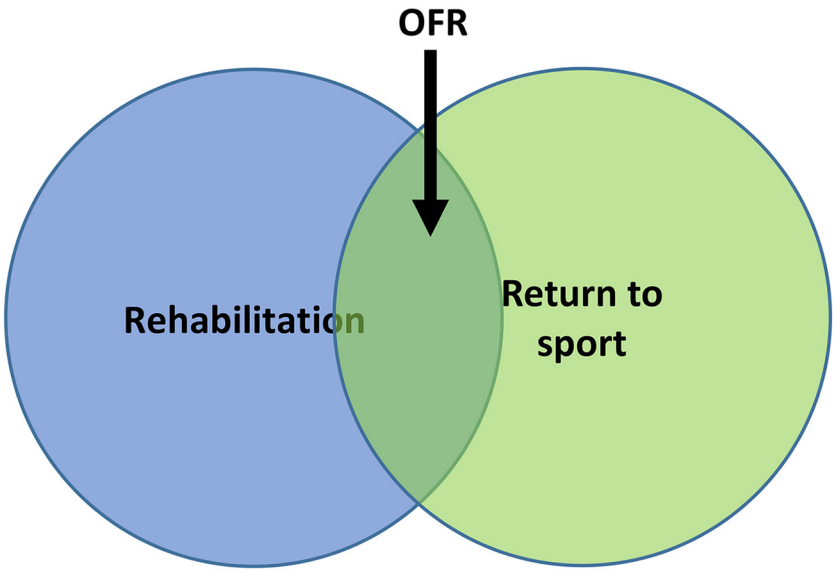


**Figure 1** A model showing the overcoming of the dichotomous conception of functional recovery with an overlap of clinical rehabilitation and return to sport: the on-field rehabilitation (OFR).

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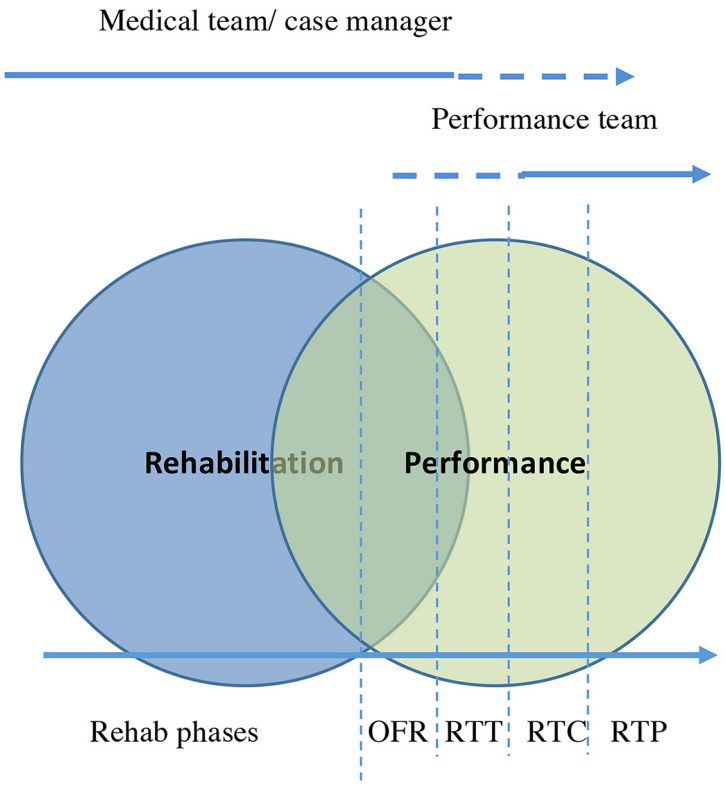


**Figure 2** Functional recovery model (return to sport and performance). The transition from the rehabilitation phases of functional recovery (rehabilitation) to the actual performance is highlighted. Four stages are indicated, starting from on-field rehabilitation (OFR), to return to training (RTT), then return to competitions (RTC) and finally return to performance (RTP). The model is applicable to any type of sport and the transition from one item to the next is based on criteria rather than on time. Above the figures indicate the person/team who are essentially in charge of the case at that period of functional recovery, involving a close working relationship between medical and performance teams during the OFR to RTC stages.

physician. Finally, the more you go to the recovery of the performance, the more places and the skills you must have which assume the characteristics of the real sports environment. As such, the process requires close collaboration and communication between the rehabilitation team (specialising in the rehabilitation and RTS process) and the performance team (specialising in $RTP_{erf}$) who should know and share profoundly their skills and roles.

## THE NEED FOR MORE COMPREHENSIVE HOLISTIC CRITERIA FOR RTS

There is a substantial agreement that it is important to establish objective criteria for safe RTS, and that an extensive test protocol should be adopted,[10 11] based on the biopsychosocial model.[9] The testing process should be embedded in functional recovery for RTS and prevention of reinjury and $RTP_{erf}$. The tests that make up the protocol should investigate some aspects:

- Clinical (pain, swelling, range of motion).
- Functional (maximum and explosive strength, both specific to the joint and global measures, muscular endurance strength, body composition).
- Biomechanical (movement analysis testing).
- Psychological (fear of reinjury, psychological attitudes).
- Sport specific (ability to support volumes and work intensities in training, sport-specific physiological screening).

Additional measures of sporting performance after RTS should be recorded and monitored to ascertain if the athlete has $RTP_{erf}$ (eg, number of matches played, goals scored, assists, number of pitches, wickets, etc, depending on the sport and performance requirements). Each test requires specific testing competencies, skills and tools, so once again emphasising the importance of interdisciplinary teamwork, open communication and shared decision-making.

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